

# **EPIDEMIOLOGICAL PREVALENCE AND MARKET DYNAMICS OF SEVERE VISUAL IMPAIRMENT: A GLOBAL AND INDIAN STRATEGIC ANALYSIS**

## **EXECUTIVE SUMMARY**

The global burden of visual impairment is substantial, with an estimated 1.1 billion people living with vision loss.<sup>1</sup> Of this total, approximately 295 million individuals suffer from Moderate-to-Severe Visual Impairment (MSVI).<sup>1</sup> Within this global context, India carries a disproportionate share, housing an estimated 4.8 million people with Severe Visual Impairment (SVI) alone—a demographic segment nearly equal in size to the 4.95 million individuals classified as blind in the country.<sup>2</sup>

SVI is functionally a critical rehabilitation category that requires specialized technological intervention to maintain productivity and reduce the substantial national economic loss, which currently stands at an estimated INR 845 billion annually due to blindness and severe vision impairment.<sup>2</sup>

The market response to this need is focused on Visual Impairment Assistive Technologies (VIAT). The global VIAT market was valued at approximately \$6.11 billion in 2024 and is experiencing rapid growth (CAGR 12.5% projected for 2024–2025).<sup>3</sup> However, the most significant immediate market opportunity is found in India, where the specialized Assistive Technology (AT) market for Special Education is projected to surge from \$350 million in 2023 to \$1.5 billion by 2027, driven by a hyper-growth CAGR of 23.5%.<sup>4</sup> This explosive growth is anchored not by consumer demand, but by mandatory governmental procurement dictated by inclusive education policies (NEP 2020) and the Rights of Persons with Disabilities (RPwD) Act, 2016.<sup>4</sup> The approximately 48,811 students categorized as "Blindness" in the Indian school system represent a guaranteed, policy-mandated B2B customer base for high-end specialized aids.<sup>6</sup>

## **SECTION 1: FOUNDATION AND DEFINITIONS: ESTABLISHING THE SCOPE OF SVI**

## 1.1 Delineation of Visual Impairment Categories (WHO Standard)

To accurately quantify the population requiring advanced assistive technology, it is essential to establish precise, globally recognized classifications for visual acuity (VA). The World Health Organization (WHO) framework, typically aligned with the International Classification of Diseases (ICD), defines visual impairment across a spectrum based on presenting visual acuity in the better eye.<sup>7</sup>

Severe Visual Impairment (SVI) corresponds specifically to **WHO Category 2**. This classification is defined by presenting visual acuity that is worse than 6/60 and better than 3/60.<sup>7</sup> Functionally, this range corresponds to **Severe Low Vision** in US notation, generally falling between 20/200 and 20/400.<sup>8</sup> Individuals classified under SVI possess residual vision that cannot be corrected by conventional spectacles or contact lenses alone, necessitating the use of specialized low-vision aids and advanced assistive technologies (VIAT) to maximize remaining functional sight.<sup>8</sup>

The spectrum extends from Category 0 (No or mild visual impairment, VA better than 6/18) through Category 1 (Moderate visual impairment, VA worse than 6/18 and better than 6/60).<sup>7</sup> The next step beyond SVI is formal Blindness, defined by WHO Categories 3, 4, and 5, where VA is worse than 3/60 or involves no light perception.<sup>7</sup>

## 1.2 Comparison of International Legal Frameworks and Functional Impact

The SVI category holds particular significance when compared against legal definitions. In the United States, **legal blindness** is defined as vision that cannot be corrected to better than 20/200 in the better eye, or a visual field restriction to 20 degrees or less.<sup>8</sup> This 20/200 threshold is equivalent to 6/60 Snellen notation.<sup>7</sup> Since SVI is defined as 6/60 or worse down to 3/60<sup>7</sup>, the SVI population largely overlaps with, or is immediately adjacent to, the group qualifying for legal disability status and associated governmental assistance programs in nations like the US.<sup>8</sup> In the UK, being classified as severely sight-impaired typically means having central visual acuity of less than 3/60.<sup>7</sup>

Global epidemiological reports frequently group SVI (Category 2) with Moderate Visual Impairment (Category 1) into the aggregate category of **Moderate-to-Severe Visual Impairment (MSVI)**.<sup>9</sup> While this aggregation provides a useful large-scale metric for policy formulation, it obscures a critical technical distinction: SVI, unlike Category 1, requires high-cost, specialized assistive technology—such as advanced magnification or sophisticated optical aids—while Category 1 may often be managed with simpler low-vision solutions.<sup>8</sup> By combining SVI with the much larger moderate VI population, the distinct funding imperative

for dedicated SVI rehabilitation services and high-end VIAT is often diminished in policy discussions.

The category of SVI represents the crucial intervention point for sophisticated rehabilitation technology. Since SVI is functionally a transitional category, situated immediately before the irrevocable boundary of total or legal blindness, the core purpose of specialized assistive technology in this segment is to maximize residual vision and thereby delay or prevent the patient from crossing the 3/60 threshold into total reliance on non-visual, or tactile, methods. Successfully maximizing the functional use of remaining vision defines the primary value proposition for the advanced VIAT industry.

Table 1: WHO/International Definitions of Visual Impairment and Legal Blindness

| WHO Category (ICD-10/11 Equivalent) | Visual Acuity Range (Better Eye)         | Severity Classification    | Key Equivalence                                   |
|-------------------------------------|------------------------------------------|----------------------------|---------------------------------------------------|
| Category 1                          | < 6/18 to $\geq$ 6/60 (20/70 to 20/160)  | Moderate Visual Impairment | Moderate Low Vision <sup>8</sup>                  |
| Category 2 (SVI)                    | < 6/60 to $\geq$ 3/60 (20/200 to 20/400) | Severe Visual Impairment   | Severe Low Vision <sup>8</sup>                    |
| Category 3 & 4                      | < 3/60 to No Light Perception (NLP)      | Blindness                  | US Legal Blindness (20/200 or worse) <sup>8</sup> |

## SECTION 2: GLOBAL AND NATIONAL PREVALENCE OF SEVERE VISUAL IMPAIRMENT

### 2.1 Global Epidemiological Burden and Trends

The global prevalence of vision impairment has reached alarming levels, driven largely by population growth and aging. An estimated 1.1 billion people worldwide live with some form of vision loss, including blindness.<sup>1</sup> Crucially, global health initiatives estimate that approximately 90% of this total vision loss is preventable or treatable.<sup>1</sup>

Focusing on the severe end of the spectrum, 43 million people worldwide live with total blindness, 77% of which is considered preventable or treatable.<sup>1</sup> The combined category of Moderate-to-Severe Visual Impairment (MSVI) affects 295 million people.<sup>1</sup> Data from 2015 confirmed that 217 million people had MSVI globally, with an overall distance visual impairment prevalence of 3.44%.<sup>9</sup>

The demographic profile of visual impairment reveals significant trends:

1. **Age:** The risk of most vision conditions increases significantly with age. As a result, 80% of visually impaired people worldwide are aged 50 years or older.<sup>9</sup>

2. **Gender:** Women account for 55% (139 million) of the world's visually impaired population.<sup>9</sup> This gender imbalance is attributed to women's longer life expectancy, which increases their exposure to age-related conditions, and in many regions, socio-economic and cultural factors that restrict women's access to preventative and curative eye health services.<sup>9</sup>
3. **Geography and Socio-Economic Status:** A striking 89% of all visually impaired people reside in low- and middle-income countries.<sup>9</sup> This burden is geographically concentrated, with three Asian regions—South Asia (73 million), East Asia (59 million), and Southeast Asia (24 million)—together housing 62% of the world's visually impaired population, despite accounting for only 51% of the world's total population.<sup>9</sup>

## 2.2 Detailed SVI Prevalence and Burden in India

India, home to more than 17% of the world's population, bears an estimated 25% of the global burden of blindness and visual impairment.<sup>11</sup> This demonstrates a disproportionate concentration of the global vision health challenge within the nation. The prevalence of vision impairment and blindness is strongly correlated with marginalized groups and lower socioeconomic status within India.<sup>11</sup>

Based on 2020 national estimates for a total population of 1.38 billion, the analysis of the different categories of impairment is possible:

- The overall prevalence of blindness (VA  $\leq$  3/60 $\times$ ) is 0.36%, translating to **4.95 million** blind persons.<sup>2</sup>
- The prevalence of Severe Vision Impairment (SVI, WHO Category 2) is 0.35%, quantifying this population at approximately **4.8 million persons**.<sup>2</sup>
- The prevalence of Moderate Vision Impairment (WHO Category 1) is 1.84%, equating to 25 million persons.<sup>2</sup>

This granular Indian data reveals a critical dynamic: the SVI population (4.8 million) is nearly equivalent in size to the strictly blind population (4.95 million).<sup>2</sup> This finding confirms that SVI is not a marginal subset but a primary demographic segment of equivalent weight to blindness. Strategically, maximizing the residual vision of these 4.8 million individuals through targeted assistive technologies offers the highest value intervention, as it sustains productivity and prevents further functional decline.

The severe visual burden in India imposes a crippling economic consequence. The estimated net loss of Gross National Income (GNI) due to blindness alone is calculated at INR 845 billion (Int\$ 38.4 billion), with a per capita GNI loss of INR 170,624 (Int\$ 7,756) per blind person.<sup>2</sup> Mitigating the progression of SVI to blindness directly addresses this massive economic strain.

Table 2: Estimated Global and Indian Population Burden (Latest Available Data)

| Impairment Category | Global Estimate | India Estimate | Source Year |
|---------------------|-----------------|----------------|-------------|
|---------------------|-----------------|----------------|-------------|

|                                       | (Millions)             | (Millions)        |                                          |
|---------------------------------------|------------------------|-------------------|------------------------------------------|
| Total Vision Loss (including Mild)    | 1,100                  | 70                | 2021 (Global), 2020 (India) <sup>1</sup> |
| Moderate to Severe VI (MSVI)          | 295                    | 29.8 (Calculated) | 2021 (Global), 2020 (India) <sup>1</sup> |
| <b>Severe Visual Impairment (SVI)</b> | N/A (Included in MSVI) | <b>4.8</b>        | 2020 <sup>2</sup>                        |
| Blindness (VA < 3/60)                 | 43                     | 4.95              | 2021 (Global), 2020 (India) <sup>1</sup> |

## SECTION 3: SVI STUDENTS AND EDUCATIONAL INCLUSION IN INDIA

### 3.1 Analysis of National Education Data (AISES/NCERT)

The educational sector in India provides quantifiable data regarding the identified population of children requiring specialized support, offering a direct metric for institutional demand for assistive technologies. Data compiled from the Concise Report of the 8th All India School Education Survey (AISES) conducted by the NCERT provides a detailed breakdown of enrolled students with disabilities.<sup>6</sup>

Visual impairment constitutes a major category within special needs education. The total number of students with visual impairments, categorized as either "Blindness" or "Low Vision," amounts to **334,670 students**, representing 15.8% of all Children with Special Needs (CWSN) enrolled in Indian schools.<sup>6</sup>

Focusing specifically on the segment requiring maximum technological adaptation:

- The category "Blindness," which encompasses students with SVI and total blindness (WHO Category 2 and above), accounts for **48,811 students** (2.3% of all CWSN).<sup>6</sup> This small but distinct segment is functionally dependent on non-visual interfaces (Braille, screen readers) or highly advanced vision enhancement technologies, making them the primary institutional consumers of specialized AT hardware and software.
- The "Low Vision" category, corresponding generally to Moderate Visual Impairment (Category 1), is significantly larger at **285,859 students** (13.5% of CWSN).<sup>6</sup> This group primarily requires magnification aids, accessible large-print materials, and basic low-vision devices.

Historically, the enrollment of visually challenged children showed an increase of 17.36% between 2002 and 2009, potentially reflecting improved awareness and increased health check-ups in schools.<sup>12</sup>

Table 3: Enrollment of Visually Impaired Students in Indian Schools (AISES Data)

| Category (AISES/NCERT)  | Total Students Enrolled | Percentage of All CWSN | Implied Severity (WHO Equivalent)                         |
|-------------------------|-------------------------|------------------------|-----------------------------------------------------------|
| Blindness               | 48,811                  | 2.3%                   | Severe VI (SVI) & Blindness (Categories 2-5) <sup>6</sup> |
| Low Vision              | 285,859                 | 13.5%                  | Moderate VI (Category 1) <sup>6</sup>                     |
| Total Visual Impairment | 334,670                 | 15.8%                  | VI/SVI/Blindness Spectrum <sup>6</sup>                    |

## 3.2 Policy Drivers and The Latent Market

The population of 48,811 students classified as "Blindness" serves as a foundational anchor for the specialized AT market in India. The government's commitment to inclusive education, enshrined in the RPwD Act, 2016, and amplified by the National Education Policy (NEP) 2020, legally compels educational institutions—both public and private—to ensure accessibility and procurement of necessary assistive tools, such as screen readers and refreshable Braille displays.<sup>4</sup> This policy mandate guarantees a stable, institutional B2B demand for specialized VIAT, regardless of immediate consumer purchasing power.

Despite these mandates, the implementation gap remains substantial: only 21.11% of the country's nearly 1.3 million schools adhere to inclusive education mandates.<sup>12</sup> This shortfall indicates a massive, impending institutional demand for AT as policy enforcement strengthens.

Furthermore, existing enrollment data likely represents a significant undercounting of the true visual impairment burden in the student population. Findings suggest that visual impairment and blindness are more common among marginalized groups.<sup>11</sup> The prevalence of visual impairment appears higher in urban schools (5.6%) compared to rural schools (3.6%), a disparity often attributed to better diagnostic access in urban areas.<sup>13</sup> Since the most marginalized populations—who are also the most likely to experience VI/blindness due to avoidable causes—are concentrated in rural and lower socioeconomic settings, widespread implementation of school-based and community-based vision screening programs is expected to trigger a mass identification event. Such an event would significantly expand the identified student population, rapidly increasing the latent demand beyond the current 334,670 reported students, thereby driving exponential growth in the educational AT sector.

## SECTION 4: THE GLOBAL ASSISTIVE TECHNOLOGY MARKET FOR VISUAL IMPAIRMENT

## 4.1 Market Sizing and Segmentation

The market for visual impairment solutions is a dynamic sector driven by medical advancements, demographic shifts, and rapid technological innovation.

The broader **Global Vision Impairment Market**, which includes medical treatments, devices, and surgery, was valued at \$6.06 billion in 2024. This market is projected to reach \$11.72 billion by 2032, reflecting a Compound Annual Growth Rate (CAGR) of 8.60% across the forecast period.<sup>14</sup>

The specific sub-segment of **Assistive Technologies for the Visually Impaired (VIAT)** represents a highly specialized and rapidly accelerating market. This market sub-segment alone was estimated at \$6.11 billion in 2024, with a projected growth to \$6.87 billion in 2025, indicating a high short-term CAGR of 12.5%.<sup>3</sup> This growth is catalyzed by the confluence of an aging population, rising awareness of assistive technologies, and increasing government initiatives globally.<sup>3</sup>

Table 4: Global Market Size for Visual Impairment Solutions (VIAS) and Assistive Technology (VIAT)

| Market Metric                               | 2024 Value (USD Billion) | Projected Value (USD Billion) | CAGR (%) | Timeframe | Scope                         |
|---------------------------------------------|--------------------------|-------------------------------|----------|-----------|-------------------------------|
| VI Solutions (Medication, Devices, Surgery) | 6.06                     | 11.72                         | 8.60%    | 2025-2032 | Broad VI Market <sup>14</sup> |
| VI Assistive Technologies (Devices only)    | 6.11                     | 6.87                          | 12.5%    | 2024-2025 | Narrow AT Market <sup>3</sup> |

The geographic distribution of the VIAT market exhibits a profound disconnect between market value and population need. While Asia harbors 62% of the global visually impaired population <sup>9</sup>, the market is currently dominated by North America (36%) and Europe (29%). Asia-Pacific accounts for only 24% of the market share.<sup>15</sup> This vast penetration gap highlights that the primary global opportunity for future, high-volume, scalable growth in VIAT lies in shifting focus toward developing and delivering affordable technologies designed for emerging economies, rather than relying exclusively on high-margin sales in established Western markets.

## 4.2 Technological Innovation and Future Trends

The future growth of VIAT is intrinsically linked to advancements in smart, portable technology. Key technological trends driving the market include:

- A projected 46% surge in AI-powered products.<sup>15</sup>
- A 39% rise in demand for portable devices.<sup>15</sup>
- Increased integration with cloud computing (34%) and wearable technology usage (28%).<sup>15</sup>

Specific high-potential segments include AI-enabled wearables and smart glasses.<sup>16</sup> Prominent market players span both hardware and software, including VFO Group, Humanware, OrCam Technologies, and specialized software providers like Jaws Screen Reader, demonstrating a reliance on sophisticated hardware (advanced magnification) and intelligent software integration (screen reading, object recognition).<sup>15</sup>

An analysis of vision loss trends indicates that while age-standardized prevalence of total blindness related to conditions like glaucoma has decreased, the prevalence of Moderate-to-Severe Visual Impairment (MSVI) due to the same conditions has increased.<sup>10</sup> This indicates a systemic shift where medical interventions are increasingly successful in managing chronic eye diseases, preventing total blindness, but often halting progression at the SVI stage. Consequently, the SVI population (WHO Category 2) represents the fastest-growing functional segment, demanding high-functioning VIAT to maximize the extended window of residual vision and maintain long-term quality of life and productivity.

## SECTION 5: STRATEGIC MARKET OPPORTUNITY IN INDIA

### 5.1 Size and Hyper-Growth in the Indian AT Market

The overall Assistive Technology market in India was valued at \$2.40 billion in 2024, forecast to grow to \$3.50 billion by 2033, exhibiting a general CAGR of 3.70%.<sup>18</sup> This moderate growth is fueled by factors such as government initiatives, demographic shifts, and increased disability awareness.<sup>18</sup>

However, the segment catering to inclusive education and disability-specific technological solutions is experiencing a period of hyper-growth. The market for **Assistive Technology and AI Solutions for Special Education** was valued at \$350 million in 2023 and is expected to rapidly escalate to \$1.5 billion by 2027.<sup>4</sup> This translates to a massive projected CAGR of 23.5%.<sup>4</sup>

Table 5: Indian Assistive Technology Market and Key Sub-Segments

| Market Scope | 2024 Value | Projected | CAGR (%) | Timeframe | Drivers |
|--------------|------------|-----------|----------|-----------|---------|
|--------------|------------|-----------|----------|-----------|---------|



|                                     | (USD Billion) | Value (USD Billion) |       |           |                                                    |
|-------------------------------------|---------------|---------------------|-------|-----------|----------------------------------------------------|
| Overall Indian Assistive Technology | 2.40          | 3.50                | 3.70% | 2025-2033 | Government initiatives, demographics <sup>18</sup> |
| Special Education AT/AI Solutions   | 0.35 (2023)   | 1.50                | 23.5% | 2023-2027 | NEP 2020, RPwD Act <sup>4</sup>                    |

The stark contrast between the general AT market's 3.7% growth and the education AT sub-market's 23.5% surge highlights a crucial strategic dynamic: governmental and institutional procurement is the immediate and dominant revenue driver for specialized VIAT in India. This indicates that market success currently hinges on B2B sales cycles, regulatory compliance, and aligning product development with the mandates of inclusive education policy, rather than relying primarily on nascent direct-to-consumer demand.

## 5.2 Policy and Regulatory Catalysts

Government policy provides the foundational structure for market demand. The Rights of Persons with Disabilities (RPwD) Act of 2016 marked a paradigm shift, mandating that all government websites and digital platforms must be accessible to persons with disabilities.<sup>5</sup> This policy, combined with the emphasis on technology in the NEP 2020, forces institutional demand for essential aids like text-to-speech tools and Braille devices.<sup>4</sup>

Further regulatory clarification is provided by the Assistive Technology (Standards and Accessibility) Rules, 2025. These rules formally classify Assistive Technology into categories, including **Specialised Assistive Technology**—defined to include devices required for specific groups, such as screen readers and augmentative communication devices—which are critical for the SVI population.<sup>19</sup> Importantly, the rules mandate that all assistive technology conform to standards prescribed by the Bureau of Indian Standards (BIS) and undergo certification before public use.<sup>19</sup>

This regulatory requirement introduces market friction but also creates a significant competitive advantage. Companies that achieve compliance and secure BIS certification for their Specialized Assistive Technology first will establish a substantial market moat, positioning them favorably to capture large-scale public tenders driven by the RPwD Act implementation.

## 5.3 Indian Innovation Ecosystem and Localization

Given that visual impairment in India is strongly associated with lower socioeconomic status<sup>11</sup> and that the majority of visually impaired individuals reside in developing economies<sup>9</sup>,

affordability and localized design are paramount. Global technology, often developed for high-income consumer bases, is frequently insufficient due to high cost and lack of localized contextual functionality (e.g., non-standardized environments, diverse languages).

The Indian market is responding through indigenous innovation. Local startups such as Innovision, TorchIt, and EYE-D are focused on leveraging Artificial Intelligence (AI) and Machine Learning (ML) to develop accessible solutions for identification, navigation, and learning.<sup>20</sup> TorchIt, for instance, focuses on state-of-the-art technological innovations to empower persons with disabilities to lead independent lives.<sup>20</sup>

The necessity of utilizing AI/ML in the Indian context is clear. Successful visual impairment solutions in a diverse environment must integrate sophisticated technology for localized text recognition across multiple scripts and languages, contextualized navigation that handles non-standardized signage and infrastructure, and reliable object recognition. AI is thus an essential feature for scalability and effectiveness across the diverse urban and rural settings of India, rather than a mere technological enhancement. This focus on local manufacturing and a low-cost product portfolio, often facilitated by collaborations with NGOs and government bodies, is the pathway to reaching the vast, underserved population.<sup>16</sup>

## **SECTION 6: STRATEGIC IMPLICATIONS AND FUTURE OUTLOOK**

### **6.1 Synthesis: Bridging Prevalence Needs with Market Solutions**

The analysis confirms that Severe Visual Impairment (SVI) requires specific, high-tech intervention strategies distinct from both moderate impairment and total blindness. The SVI population in India (4.8 million) is a strategically critical cohort.<sup>2</sup> Investment in assistive technologies for this group provides the maximum leverage in mitigating the national economic burden and maximizing functional independence, especially considering that medical advances are increasingly preventing progression to complete blindness, thereby stabilizing the SVI population as a prime long-term market segment.<sup>10</sup>

The education sector, while representing a small fraction of the total SVI population (48,811 students in the "Blindness" category), acts as the critical market engine.<sup>6</sup> Inclusive policies (RPwD Act, NEP 2020) transform this demographic into a guaranteed institutional B2B procurement pipeline. Technologies validated and standardized within this education framework can then be scaled to address the remaining 4.75 million adults with SVI in the general population.

### **6.2 Recommendations for Investment and Policy**

1. **Prioritize B2B Strategy for Education:** VIAT manufacturers seeking entry or expansion in India must prioritize institutional B2B sales cycles over direct-to-consumer models in the immediate term. Product development and distribution must align with the procurement needs of schools and government agencies, focusing on specialized items like certified Braille devices, high-quality screen readers, and adaptive learning platforms.<sup>4</sup>
2. **Focus on Regulatory Compliance as a Competitive Edge:** Investment in research and development should emphasize adherence to the forthcoming Bureau of Indian Standards (BIS) certification requirements for Specialised Assistive Technology.<sup>19</sup> Gaining early certification will secure favorable positioning in government tenders mandated by the RPwD Act.
3. **Invest in Localized AI Solutions:** To address the vast and underserved rural market and overcome linguistic and infrastructural heterogeneity, technology investment should target Indian startups specializing in low-cost, multi-lingual, and multi-functional SVI devices leveraging AI and ML for contextual awareness and text processing.<sup>20</sup>
4. **Advocate for Systematic Screening Programs:** Policy efforts must aggressively push for the expansion of rural vision screening programs. The current educational enrollment data, showing lower VI prevalence in rural areas, likely masks a huge latent demand.<sup>13</sup> Identification of this underdiagnosed population will validate the need for massive subsequent infrastructure and AT procurement budgets.

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